

AX88179A Product Introduction

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Revision History

Revision	Date	Description
1.00	2022/06/15	Initial release
1.01	2022/07/06	Updated Figure 3 in Section 3.
1.02	2022/12/27	Updated Figure 1 in Section 2.
1.03	2023/08/15	Updated Figure 3 in Section 3.
1.04	2024/01/17	Updated Selection Guide in Figure 3.
1.05	2024/05/13	Updated Selection Guide in Figure 3.

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1. Introduction

The AX88179A USB 3.2 Gen1 to 10/100/1000M Gigabit Ethernet controller is a high performance and highly integrated ASIC which enables low cost, small form factor, and simple plug-and-play Gigabit Ethernet network connection capability for desktops, notebook PC's, Ultrabook's, docking stations, game consoles, digital-home appliances, and any embedded system using a standard USB port.

The AX88179A features a USB interface to communicate with a USB Host Controller and is Compliant with USB specification V3.2 Gen1, V2.0, and V1.1. It implements a 10/100/1000Mbps Ethernet LAN function based on IEEE802.3, IEEE802.3u, and IEEE802.3ab tandards with embedded SRAMs for packet buffering. And, it also integrates an on-chip 10/100/1000Mbps EEE-compliant Ethernet PHY to simplify system design.

This document provides an overview of AX88179A USB 3.2 Gen1 to 10/100/1000M Gigabit Ethernet controller product.

2. Block Diagram

The following is AX88179A block diagram,

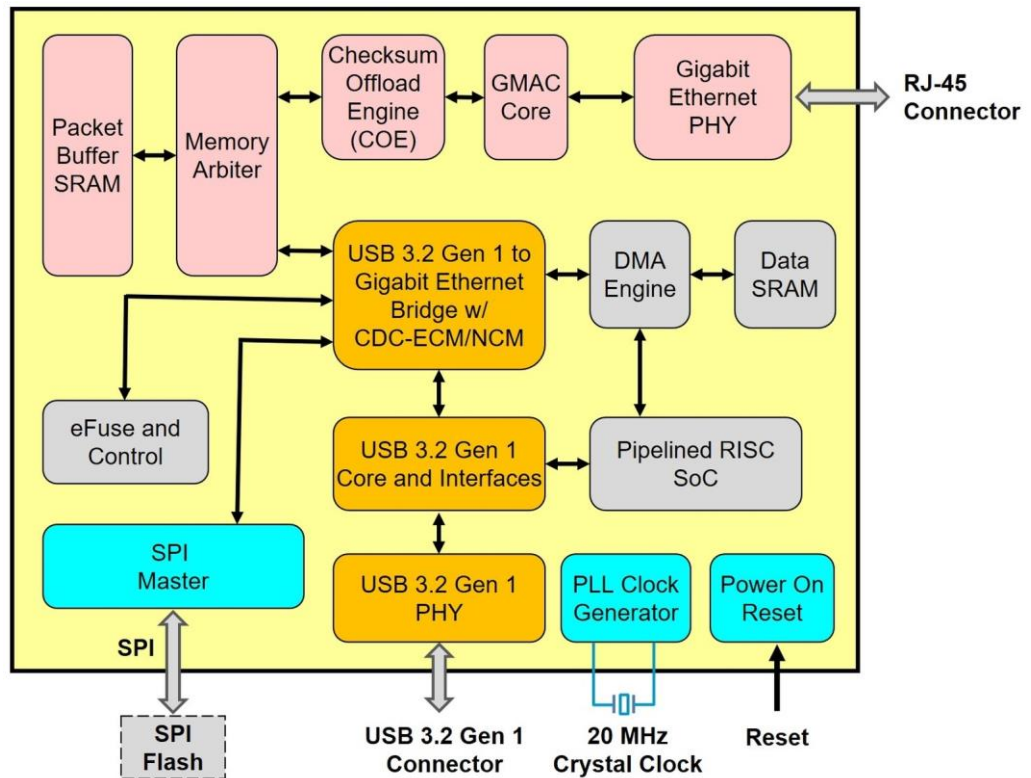


Figure 1. AX88179A Block Diagram

2-1. Typical System Block Diagram



Figure 2. AX88179A Typical Applications Diagram

3. Selection Guide

The following is the selection guide of ASIX Electronics USB Ethernet ICs for different requirement applications. Please visit ASIX Electronics USB Ethernet ICs web pages and contact ASIX's Sales (sales@asix.com.tw) for details.

Part No.	USB Speed	Ethernet MAC/ PHY (bps)	CDC-NCM Driverless	iPXE	PTP	MAC Interface	Temperature Range (°C)	Package
AX88279	USB 3.2	2.5G/1G/100M	V	V	V	-	0 ~ +70	QFN-60
AX88179B	USB 3.2	1G/100M/10M	V	V	V	-	0 ~ +70 -40 ~ +85	QFN-40
AX88179A	USB 3.2	1G/100M/10M	V	V	-	-	0 ~ +70	QFN-40
AX88772E	USB 2.0	100M/10M	V	V	V	-	0 ~ +70	QFN-32
AX88772D	USB 2.0	100M/10M	V	V	-	-	0 ~ +70	QFN-32
AX88772C	USB 2.0	100M/10M	-	-	-	RMII/ Rev-RMII	0 ~ +70/ -40 ~ +85	LQFP-64
AX88772B	USB 2.0	100M/10M	-	-	-	RMII/ Rev-RMII	0 ~ +70/ -40 ~ +85	LQFP-64
AX88772A	USB 2.0	100M/10M	-	-	-	-	0 ~ +70	LQFP-64

Figure 3. Selection Guide

4. Ordering Information

The following are the ordering information of AX88179A silicon and AX88179A demo board. Please contact ASIX's Sales (sales@asix.com.tw) for more details.

Part Number	Description
AX88179AQF	40-pin QFN lead Free package, Commercial temperature range: 0 to 70°C.

AX88179A Demo Board	Description
AX88179A Demo Board	This is a USB dongle for AX88179A USB 3.2 to Gigabit Ethernet application

Figure 4. Ordering Information

5. Target Applications

The following are some PC/Internet and consumer electronics target applications for your reference.

PC/Internet



Consumer Electronics



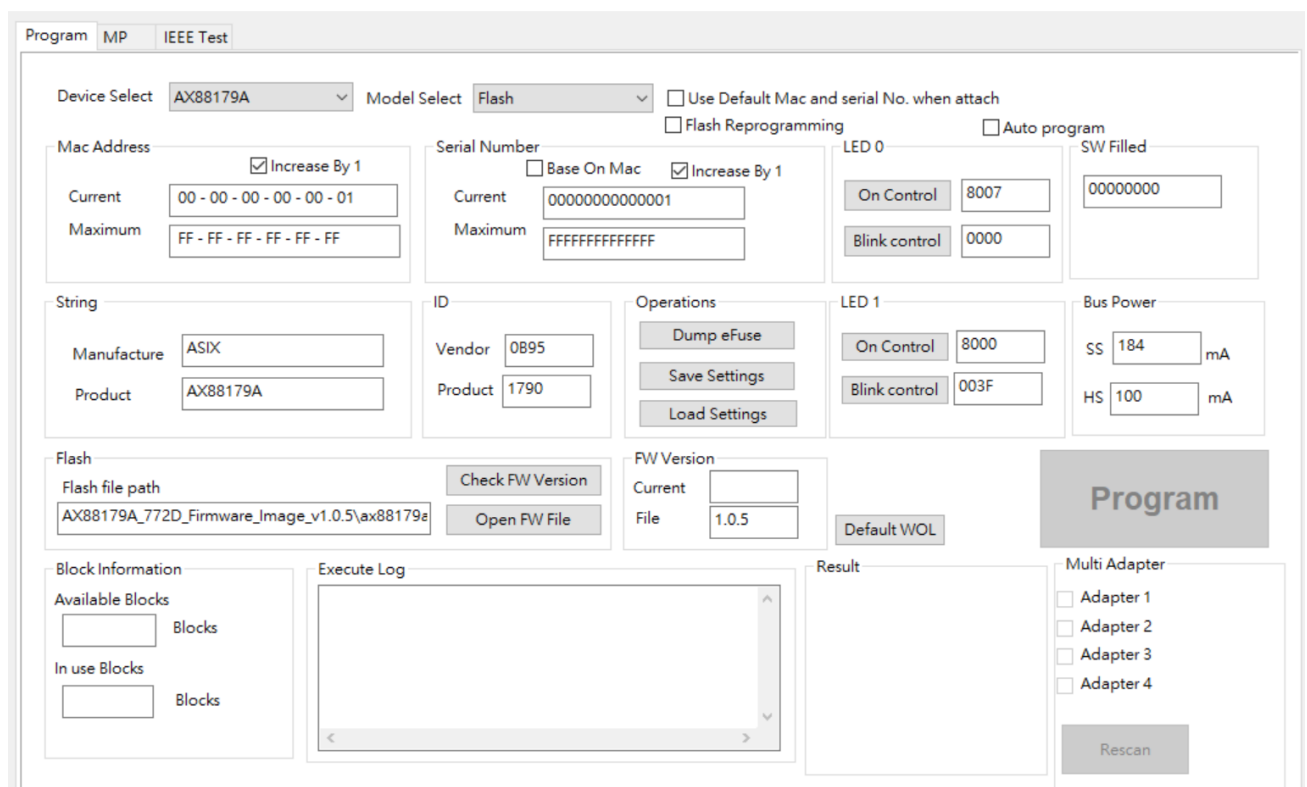
Figure 5. Target Applications

6. Mass Production Solutions

To support the mass production for those products using AX88179A chip, ASIX provides the Windows Mass Production Tool solution for AX88179A customers. This chapter provides a brief introduction for this solution. Please refer to “**AX88179A Mass Production User Guide**” for details.

6-1. Windows Mass Production Tool

ASIX Electronics provides a Windows Mass Production tool for users to easily program the Serial Flash/eFuse of AX88179A on a typical Windows PC. This AX88179A Windows Mass Production Tool supports to customize the MAC address, Serial Number, Vendor ID and Product ID, etc. for AX88179A based application systems in mass production.



The screenshot displays the 'Program' page of the Windows Mass Production Tool. The interface is organized into several sections:

- Top Navigation:** Tabs for 'Program', 'MP', and 'IEEE Test'.
- Device and Model Selection:** 'Device Select' is set to 'AX88179A' and 'Model Select' is set to 'Flash'. There are checkboxes for 'Use Default Mac and serial No. when attach' and 'Flash Reprogramming'.
- Mac Address Configuration:** Includes 'Current' (00-00-00-00-00-01) and 'Maximum' (FF-FF-FF-FF-FF-FF) fields, with an 'Increase By 1' checkbox.
- Serial Number Configuration:** Includes 'Current' (0000000000000001) and 'Maximum' (FFFFFFFFFFFFFF) fields, with 'Base On Mac' and 'Increase By 1' checkboxes.
- String Configuration:** Fields for 'Manufacture' (ASIX) and 'Product' (AX88179A).
- ID Configuration:** Fields for 'Vendor' (0895) and 'Product' (1790).
- Operations:** Buttons for 'Dump eFuse', 'Save Settings', and 'Load Settings'.
- LED Control:** Sections for 'LED 0' and 'LED 1' with 'On Control' and 'Blink control' fields.
- Bus Power:** Fields for 'SS' (184 mA) and 'HS' (100 mA).
- Flash Configuration:** 'Flash file path' (AX88179A_772D_Firmware_Image_v1.0.5\ax88179a), 'Check FW Version', and 'Open FW File' buttons.
- FW Version:** Fields for 'Current' and 'File' (1.0.5).
- Block Information:** 'Available Blocks' and 'In use Blocks' sections.
- Execute Log:** A text area for logging execution details.
- Result:** A text area for displaying results.
- Multi Adapter:** Checkboxes for 'Adapter 1', 'Adapter 2', 'Adapter 3', and 'Adapter 4', with a 'Rescan' button.
- Program Button:** A large, prominent button to initiate the programming process.

Figure 6. Program Page for Windows Mass Production Tool

This tool also supports to send/receive packets in different Ethernet speed modes, and program the serial Flash/eFuse. This tool can be run on a Windows PC, which installs the AX88179A Windows driver. This tool also needs a separate server PC to run the test server tool (TestServer.exe). The test server tool on server PC can receive packets from the “device under test” product, and then reply back.

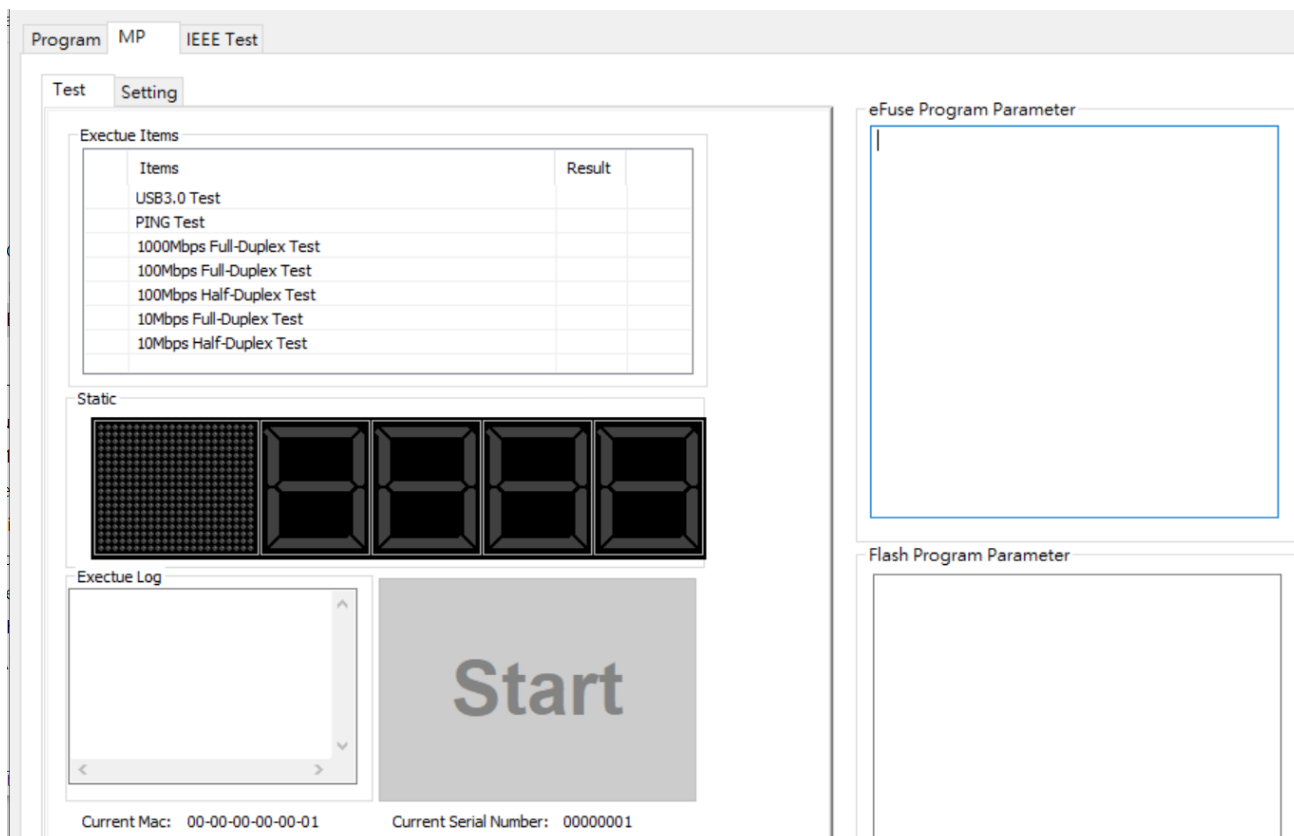


Figure 7. MP Page for Windows Mass Production Tool

7. AX88179A Demo Board

ASIX Electronics provides AX88179A demo board for users to evaluate the basic functions of AX88179A on different target applications. If you need to purchase the AX88179A demo board, please contact ASIX's Sales (sales@asix.com.tw) for more details.



Figure 8. AX88179A Demo Board



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